Hexadecimal to Binary & Binary to Hexadecimal Conversion

$r = 16$

» In hexadecimal number system we have 16 different digits.
» Each one of them can be represented by a equivalent 4-bit binary number.

0 to (16-1)
0 to 15

	0	-	0000
	1	-	0001
	2	-	0010
	3	-	0011
	4	-	0100
	5	-	0101
	6	-	0110
	7	-	0111
	8	-	1000
	9	-	1001
10	A	-	1010
11	B	-	1011
12	C	-	1100
13	D	-	1101
14	E	-	1110
15	F	-	1111

Hex to Binary :-

i) $(259A)_{16} \longrightarrow (0010\ 0101\ 1001\ 1010)_2$

0010 0101 1001 1010

ii) $(CAFE.3D)_{16} \longrightarrow (1100\ 1010\ 1111\ 1110.0011\ 1101)_2$

1100 1010 1111 1110 . 0011 1101

11 B - 1011
12 C - 1100
13 D - 1101
14 E - 1110
15 F - 1111

Binary to Hex :-

i) $(\underbrace{0001}_{1}\ 1000\ 1001\ .\ 1100)_2 \longrightarrow (189.C)_{16}$

Hw i) $(FACE)_{16} \longrightarrow (\qquad\qquad)_2$

ii) $(1101\ 1110\ 1010\ 1111)_2 \longrightarrow (\qquad\qquad)_{16}$

1. 1111 1010 1100 1110
2. DEAF